

SESSION 1 TASK

1. List three apps you use daily (such as Zomato, Instagram, or Flipkart) and describe one feature in each app that likely uses machine learning. For each, state whether it is an example of supervised, unsupervised, or reinforcement learning.

→ (i) Instagram

Feature: Reels and post recommendation (Explore page)

How ML is used: It suggests videos/posts based on what you like, watch, and interact with.

Type of Learning: Reinforcement Learning + Supervised Learning

- Reinforcement: learns from your clicks, likes, watch time (feedback loop)
- Supervised: trained on labeled user interaction data

(ii) ChatGPT (AI Assistant)

Feature: Answer generation / conversational responses

How ML is used: I generate answers based on patterns learned from large text datasets and user prompts. I also improve responses based on feedback signals (helpful / not helpful).

Type of Learning:

- Supervised Learning: trained on large datasets of text with expected outputs
- Reinforcement Learning from Human Feedback (RLHF): improves responses based on human ratings and preferences

(iii) Zomato

Feature: Restaurant recommendations

How ML is used: It suggests restaurants based on your location, order history, and ratings.

Type of Learning: Supervised Learning

- Uses past labeled data like ratings, orders, and reviews to predict what you may like.

2. Given the following scenarios, classify each as supervised, unsupervised, or reinforcement learning: 1) Netflix recommending movies based on your watch history, 2) Spotify grouping similar songs into playlists, 3) A self-driving car learning to park by trial and error. Write your answers with a one-line explanation for each choice.

→ 1) Netflix recommending movies based on your watch history:

Type: Supervised Learning

Explanation: It uses past labeled user behavior (watched/liked movies) to predict what you may like next.

2) Spotify grouping similar songs into playlists:

Type: Unsupervised Learning

Explanation: It groups songs based on similarity in features like mood, tempo, and genre without labeled outputs.

3) Self-driving car learning to park by trial and error:

Type: Reinforcement Learning

Explanation: The car learns by interacting with the environment and improving based on rewards (successful parking) or penalties (errors).

3. Draw a simple diagram (hand-drawn or digital) showing the basic ML workflow: Data → Preprocessing → Modeling → Evaluation → Deployment. Under each step, write a one-line example of what happens at that stage for a food delivery app like Swiggy.

→ 1. Data

What happens: Raw data is collected.

Swiggy example:

User order history, restaurant ratings, delivery time, location data.

2. Preprocessing

What happens: Data is cleaned and prepared.

Swiggy example:

Removing missing ratings, encoding restaurant categories, handling null delivery times.

3. Modeling

What happens: Machine learning model is trained.

Swiggy example:

The model learns to predict which restaurants a user will likely order from.

4. Evaluation

What happens: Model performance is tested.

Swiggy example:

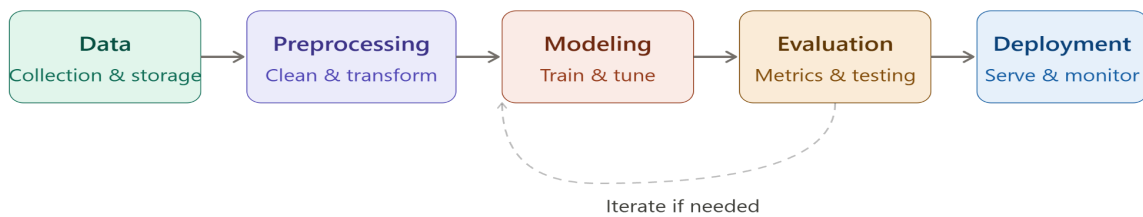
Checking accuracy of predicted food recommendations using test user data.

5. Deployment

What happens: The model is used in real application.

Swiggy example:

Recommendations appear in the Swiggy app like "You may also like these restaurants".



4. Imagine you are building a spam filter for WhatsApp messages. Write a short paragraph explaining how traditional programming would approach this problem versus how machine learning would approach it.
Hint: Focus on rules vs learning from data.

→ Traditional Programming vs Machine Learning (Spam Filter)

In traditional programming, a WhatsApp spam filter would be built using fixed rules written manually by humans, such as blocking messages containing words like “win money”, “free offer”, or repeated links. However, this approach is limited because spammers can easily change wording and bypass rules.

In machine learning, the system learns from large amounts of labeled data containing spam and non-spam messages. Instead of fixed rules, the model identifies patterns (like message frequency, links, keywords, and sender behavior) and automatically improves its accuracy over time, making it more flexible and effective against new types of spam.